

MCT-454L Mobile Robotics

Lab 7: Autonomous Navigation with Nav2 and Multi-Waypoint Mission Planning

1. Objective

The objective of this lab was to implement a full autonomous navigation pipeline in ROS 2. This involved utilizing a static map generated in Lab 5, localizing the robot using Adaptive Monte Carlo Localization (AMCL), and executing multi-waypoint missions using the Nav2 (Navigation 2) stack.

2. Task 1: Single Goal Navigation

Using the "Nav2 Goal" tool in RViz, the robot was assigned individual destinations. The following table tracks the performance of the AMCL localization and the Nav2 planner.

Goal #	Start Pose (X, Y)	End Pose (X, Y)	Result	Observation
1	0.00, 0.00	1.25, -0.45	Success	Smooth global path calculated.
2	1.25, -0.45	-1.10, 0.85	Success	Navigated between pillars without collision.
3	-1.10, 0.85	0.05, 0.02	Success	Returned to origin with high localization accuracy.

3. Task 2: Multi-Waypoint Mission Planning

A sequence of 5 waypoints was defined to create a "safe square" patrol route within the central area of the TurtleBot3 world. These coordinates were hardcoded into the Python node to satisfy Task 2.

Waypoint Table:

Waypoint #	X (m)	Y (m)	Z (m)	Yaw (W)	Description
1	0.5	-0.5	0.0	1.0	Inner Bottom-Right
2	-0.5	-0.5	0.0	1.0	Inner Bottom-Left
3	-0.5	0.5	0.0	1.0	Inner Top-Left
4	0.5	0.5	0.0	1.0	Inner Top-Right
5	0.0	0.0	0.0	1.0	Return to Origin

4. Task 3: Dynamic Waypoint Injection

To demonstrate dynamic mission flexibility, the `waypoint_navigator` node was executed using command-line arguments. This allows the user to specify coordinates without modifying the source code.

Example Command:

```
ros2 run lab7_nav waypoint_navigator 0.5 0.0 1.0 -0.5 0.0 1.0 0.0 0.0 1.0
```

5. Task 4: Costmap Observation

Using the `ros2 topic list | grep costmap` command, the following critical topics were identified:

- `/global_costmap/costmap`
- `/local_costmap/costmap`

Analysis of Roles:

- **Global Costmap:** Responsible for macroscopic path planning. It uses the static map to create an "inflation layer" around known walls, ensuring the global planner picks a path that maintains a safe distance from permanent structures.
- **Local Costmap:** Responsible for reactive, microscopic avoidance. It uses live LiDAR data to detect dynamic obstacles (like the box added in Task 5) that are not on the original map, allowing the robot to deviate from the global path to stay safe locally.

6. Task 5: Navigation Recovery Behaviors

During the mission, a box was dropped into the robot's path in Gazebo to force a recovery state.

Observations:

1. **Detection:** The local costmap immediately registered the box as a high-cost obstacle.
2. **Recovery Count:** The robot attempted multiple recoveries (12+) because the inflation radius was large, making the gap around the box appear impassable.
3. **Behavior:** The robot executed "Spin" and "Wait" behaviors.
4. **Handler:** Using `rqt_graph`, the `/behavior_server` was identified as the node managing these emergency actions.

7. Conclusion

This lab successfully integrated the mapping, localization, and planning layers of the ROS 2 ecosystem. The transition from Lab 5 (Mapping) to Lab 7 (Navigation) highlighted that a map is only a reference; true autonomy requires the real-time interaction between AMCL localization and the Nav2 costmap servers to handle dynamic environments. The primary challenge was tuning waypoint coordinates to avoid the inflation layers of the pillars.